



Prenatal and Early Childhood Ambient Heat Exposure and Developmental Outcomes: A Systematic Review

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Accepted: 12 April 2026

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Abstract

Record-high temperatures and more frequent heatwaves are well-documented consequences of climate change, but the developmental consequences of heat during sensitive periods such as pregnancy and early childhood are unclear. We conducted a pre-registered systematic review (PROSPERO CRD42024596655) of empirical studies examining associations of ambient heat exposure during the prenatal period or ages 0–5 with developmental outcomes measured later in childhood, adolescence, or adulthood. Eleven databases were searched in August 2024 and supplemented with backward and forward citation tracking. Two reviewers independently screened, extracted, and appraised studies using the Mixed Methods Appraisal Tool. Findings were synthesized narratively due to heterogeneity in design, exposure, and outcomes. Of 11,955 records, five studies met inclusion criteria. Four reported that higher early-life heat exposure was associated with adverse cognitive, mental health, or behavioural outcomes across the life course, while one found null effects. Effect modification was observed by developmental timing, region, season, and contextual factors. Retrieved studies were geographically narrow and methodologically heterogeneous. Although data are limited, emerging findings suggest that heat exposure in early life may have lasting adverse effects on development. More rigorous and more globally representative studies are needed to clarify mechanisms, sensitive windows, and moderators, and to inform prevention and adaptation.

Keywords Early childhood · Pregnancy · Human development · Heat · Ambient temperature · Climate change

Introduction

Record-high temperatures and increasingly frequent heat waves are among the most visible and well-documented consequences of global climate change (IPCC, 2023; Romanello

et al., 2024). A growing body of systematic reviews and meta-analyses demonstrates that excessive heat affects not only physical health, including maternal, fetal, and neonatal health (Lakhoo et al., 2025; Weeda et al., 2024), but also mental health and human behavior (Ebi et al., 2021; Kiarsi et al., 2023; Thompson et al., 2023). For instance, one meta-analysis found that a 1 °C rise in mean monthly temperature was associated with a 1.5% increase in suicide incidence and a 9.7% increase in hospital attendance or admissions for mental illness (Thompson et al., 2023). Their broader narrative review of 144 studies further revealed consistent evidence of adverse impacts on community mental health and well-being. Similarly, a meta-analysis of heat effects on children's education outcomes found that 1 °C temperature was associated with a 0.025 standard deviation decrease in standardized test performance (Prentice et al., 2025), while a meta-analysis of eight studies reported that a 10 °C increase in short-term mean temperature was linked to a 9% increase in violent crime (Choi et al., 2024). Taken together, these findings indicate that excessive heat contributes to

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significant individual and societal costs through its effects on morbidity, mortality, and social functioning.

Despite growing evidence linking heat exposure to physical, mental, and behavioral outcomes, most research has focused on adults with a small handful of studies on adolescents. Far less is known about the effects of ambient heat exposure during the prenatal and early childhood periods, and its implications for developmental outcomes across the life course (Cuartas et al., 2024; Sanson et al., 2019, 2022). Yet these are sensitive periods in which biological and psychosocial processes lay the foundation for lifelong health, learning, and well-being (Berens & Nelson, 2019; Draper et al., 2024; Vergunst & Berry, 2022). Understanding the developmental consequences of early exposure to excessive heat is therefore critical for anticipating individual and societal trajectories in a warming world.

In this study, we shed light on these issues by conducting a systematic review of studies on the links between prenatal and early childhood ambient heat exposure and developmental outcomes across the life course. For the purpose of this study, we conceptualize developmental outcomes as the relatively enduring capacities and functioning across cognitive, emotional, behavioral, and mental health domains that emerge from early biological and psychosocial processes and shape individuals' trajectories of learning, wellbeing, and adaptation across the life course. We define ambient heat as the outdoor thermal environment at a given place and time, typically indexed by air temperature (e.g., daily maximum) and, where relevant, by simple heat-stress indices such as the heat index or apparent temperature. We define ambient heat exposure as the ambient heat assigned to individuals or populations over a specified period by linking location and time specific temperature (or heat-index) data to relevant windows (e.g., prenatal trimesters, early childhood).

Ambient Heat and the Ecology of Human Development

We employ an ecological model (Bronfenbrenner & Morris, 2006) to theorize how exposure to excessive ambient heat early in life may influence developmental outcomes across the life course. Development emerges from dynamic interactions between the individual and their contexts, ranging from the most proximal microsystem (e.g., home, caregiving, neighborhood), mesosystem (e.g., links between work and family), and exosystem (e.g., parental employment conditions, community infrastructure), to the most distal macrosystem (e.g., norms, socioeconomic structures, policy environments). These systems evolve across the chronosystem, which captures historical and life course changes (e.g., economic instability) that alter developmental trajectories

(Bronfenbrenner, 2005). A central principle of this perspective is reciprocity: contexts shape individuals, and individuals actively shape their contexts.

The recently proposed geo-bioecological model, an expansion of traditional ecological perspectives, emphasizes the need to integrate and recognize the critical role of environmental and climatic factors in shaping individual development and surrounding contexts (Cuartas & Morris-Perez, 2025). Among these factors is ambient temperature, an omnipresent force that influences the establishment of human settlements, agricultural patterns, human conflicts, daily social interactions, opportunities for indoor and outdoor activities, and overall wellbeing (Koepp et al., 2023; Kohler et al., 2017; Sapolsky, 2017). Climate change, driven by historical processes such as industrialization, has accelerated global warming, resulting in sustained increases in average temperatures, as well as more frequent and severe heatwaves and climatic anomalies within a single person's lifespan (Magnan et al., 2021; Romanello et al., 2024).

Sensitive Periods of Development and Mechanisms

The developing person, often placed at the center of ecological models, may be more sensitive to environmental input at different developmental stages. In particular, the prenatal and early childhood periods are considered sensitive periods of development, during which environmental exposures can have larger and longer-lasting consequences on skill development (Cuartas et al., 2024; Vergunst & Berry, 2022). During these years, the brain and multiple biological systems (such as the neuroendocrine, cardiovascular, thermoregulatory, and metabolic systems) are especially plastic and undergo rapid adaptation to optimize the organism to its context (Berens & Nelson, 2019). With respect to ambient temperature, both the prenatal and early childhood periods represent windows of heightened vulnerability: biological systems are more sensitive to excessive heat, young children's bodies heat up more quickly and have less capacity to dissipate heat (e.g., through sweating), and they are fully dependent on caregivers to access water or cooler spaces. These factors, together amplify the potential risks of excessive heat exposure early in life (Early Childhood Scientific Council on Equity and the Environment, 2023; Vergunst & Berry, 2022).

The ecological perspective suggests that prenatal and early childhood exposure to excessive heat can affect the developing person through both direct and contextual mechanisms. Directly, excessive heat can lead to dehydration, increased heart rate, overactivation of the immune system, and even overheating of the brain, all of which can compromise biological and skill development (Early Childhood Scientific Council on Equity and the Environment,

2023). Indirectly, excessive heat can influence prenatal and early childhood outcomes through cascading effects across micro-, meso-, exo-, and macrosystems. For example, meta-analyses indicate that excessive heat may increase the risk of preterm birth, stillbirth, congenital anomalies, and other adverse gestational and fetal outcomes (Lakhoo et al., 2025), all of which are known risk factors for long-term developmental challenges (Walker et al., 2011). Extreme heat can make outdoor environments less safe or comfortable, limiting opportunities for play and learning outside the home, increasing sedentary behavior, and reducing interpersonal interactions that are essential for early development (Koepp et al., 2023).

At the family level, excessive heat can also exacerbate stress, mental health challenges, and impulsivity among mothers, fathers, and other caregivers, potentially increasing aggressive behaviors (e.g., violent punishment) and undermining the cognitive and emotional capacities they rely on to provide nurturing care (Cuartas & Camacho, 2025; Cuartas & Vergunst, 2025). It has also been associated with increases in interpersonal violence, armed conflict, and crime (Burke et al., 2015; Hsiang et al., 2013; Levy et al., 2017; Miles-Novelo & Anderson, 2022), which are well-established risk factors for poor early development (Goto et al., 2021; Kadir et al., 2019). Finally, heat can affect crops and food security, undermining adequate nutrition, a critical input for healthy development (Draper et al., 2024; Romanello et al., 2024).

The Present Study

Despite a strong conceptual rationale and evidence linking excessive heat to mechanisms that may affect human development, there is limited systematic evidence on its direct impact in early life. While numerous advocacy, conceptual, and theoretical pieces highlight the potential connections between climate change, particularly excessive heat, and human development (e.g., Cuartas et al., 2024; Early Childhood Scientific Council on Equity and the Environment, 2023; Rees, 2021; Vergunst & Berry, 2022), empirical work remains scarce. A recent rapid review identified only one study examining associations between prenatal or early childhood heat exposure and developmental outcomes (Cuartas et al., 2024). Given the timeliness of this issue for guiding research, health policy, and climate adaptation, there is an urgent need for systematic evidence to clarify what is currently known about the link between early in life heat stress and developmental outcomes.

The purpose of this study was to systematically review the association between prenatal and early childhood exposure to ambient heat and developmental outcomes, with a critical assessment of the methods, findings, and limitations

of existing studies to inform research, policy making, and climate change adaptation. Specifically, we aim to address the following research questions:

1. What is the direction and magnitude of associations between prenatal/early childhood ambient heat exposure and later developmental outcomes across the life course?
2. What contextual and individual characteristics could potentially modify the association of prenatal and early childhood ambient heat exposure with developmental outcomes?
3. What analytic approaches have been used in the literature to assess associations of prenatal and early childhood ambient heat exposure with developmental outcomes?

Methods

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA; Page et al., 2021). Supplemental file S1 presents the PRISMA 2020 checklist. The search strategy and methods were prespecified in a preregistered protocol (PROSPERO ID: CRD42024596655).

Search Strategy

The following 11 electronic databases were searched by the University of Oslo head librarian (Glenn Karlsen Bjerkenes) between August 9 and 13, 2024: (1) APA PsycINFO; (2) ERIC; (3) Education Research Complete; (4) GreenFILE; (5) Embase; (6) MEDLINE; (7) Scopus; (8) EconLit; (9) Sociological Abstracts; (10) Global Health; and (11) CINAHL. The search strategy combined keywords related to ambient heat (e.g., “temperature*,” “heat*,” “global warming”), the prenatal and early childhood periods (e.g., “prenat*,” “infan*,” “preschool*”), and developmental outcomes (e.g., “develop*,” “socioemotional,” “child behavior*”). Full search documentation, including database-specific code, is available in the supplemental file S2.

The only deviation from the pre-registered protocol was the use of backward and forward citation tracking to enhance comprehensiveness and reduce the likelihood of missing relevant studies. This included manual screening of reference lists from included studies as well as the use of *Citationchaser* (Haddaway et al., 2021), which identifies all citing records in the Lens.org database (covering PubMed, PubMed Central, CrossRef, Microsoft Academic Graph, and CORE).

Inclusion and Exclusion Criteria

We defined inclusion and exclusion criteria using the Population, Intervention/Exposure, Comparator, Outcomes (PICO) framework (Dekkers et al., 2019). We included studies that examined associations between ambient heat exposure during the prenatal period or early childhood (up to age five) and developmental outcomes measured in childhood, adolescence, or adulthood (Population and Exposure). Eligible studies assessed ambient heat, higher temperatures, heat waves, or hot weather and included a comparison group with lower exposure levels (Comparator). Outcomes of interest included composite indices of child development, cognitive and language skills, social and emotional skills, mental and behavioral health, executive function, and motor skills (Outcomes). We excluded qualitative or case studies, reviews (systematic, scoping, or meta-analyses), books, book chapters, conference papers, and other gray literature not specified in the inclusion criteria. We also excluded studies that did not assess ambient heat exposure prenatally or in early childhood, lacked a comparison group, or did not report developmental outcomes.

Study Selection and Data Extraction

We searched each database separately and imported all records to Covidence (<https://www.covidence.org/>). After excluding duplicates, two authors independently screened titles and abstracts, and then reviewed all full texts, according to inclusion criteria. Disagreements were resolved by consensus through discussion among all authors.

Two authors independently conducted data extraction using a pre-piloted template in Covidence and resolved any disagreements by consensus. We extracted the following information from each study (1) study characteristics (e.g., author and year of publication), (2) population characteristics (e.g., country, sample size, age, and sex or gender, as reported by the authors), (3) exposure characteristics (e.g., measurement approach), (4) outcome characteristics (e.g., measurement approach), (5) methodological characteristics (e.g., study design, data structure, analytic approach), (6) main findings, including effect sizes in the metric reported in the studies (7) findings from moderation analyses, if available, (8) study conclusions.

Data Synthesis and Quality Assessment

We conducted a narrative review of the evidence, critically examining methodological approaches and key findings to identify study limitations and implications for future research, policy, and practice. Although no pre-specified subgroup analyses were planned, we narratively synthesized

results from any reported moderation analyses. We used the Mixed Methods Appraisal Tool (MMAT; Hong et al., 2018) to assess basic study quality. The MMAT includes five quality criteria for quantitative, non-randomized studies: (1) Are the participants representative of the target population?; (2) Are measurements appropriate regarding both the outcome and intervention (or exposure)?; (3) Are there complete outcome data?; (4) Are the confounders accounted for in the design and analysis?; and (5) During the study period, is the intervention administered (or exposure occurred) as intended? For the purposes of this study, we operationalized the final criterion as requiring that the exposure variable assessed heat as expected. Two authors independently rated each quality criterion as yes, no, or unclear, and resolved any inconsistencies through consensus. We did not compute a summary MMAT score, consistent with guidance, but interpreted risk of bias by MMAT sub domains when synthesising results.

Regarding certainty of the evidence, we did not conduct a formal GRADE assessment because the small number of studies and substantial heterogeneity of exposures, outcomes, and study designs precluded meaningful quantitative synthesis. Instead, we assessed the overall certainty of the evidence narratively, informed by study-level risk of bias appraised with the MMAT.

Results

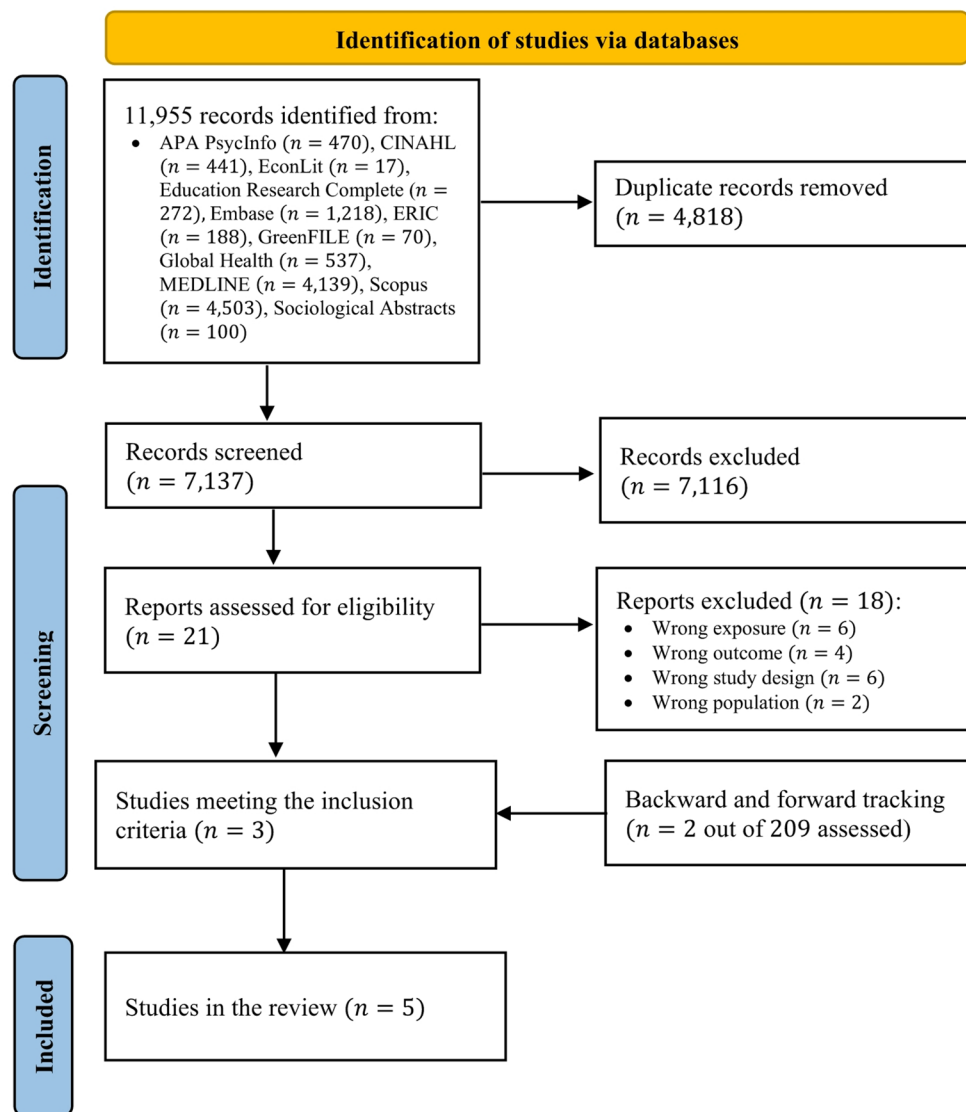
Search Results

We identified a total of 11,955 records, of which 4,818 were duplicates. After removing duplicates, 7,137 records remained. Following title and abstract screening, 7,116 records were excluded. Of the remaining 21 records, 18 were excluded because their exposure variables were not related to heat, they did not include developmental outcomes, or they otherwise failed to meet the inclusion criteria. Two additional studies meeting our inclusion criteria were identified through backward and forward tracking of 209 articles. Consequently, five studies met all inclusion criteria and were retained for analysis. (See Fig. 1 for a detailed flow diagram).

Narrative synthesis

Table 1 summarizes the sample characteristics, methods, main findings, and limitations of each included study.

The evidence base is geographically limited and heterogeneous, comprising studies from China (two studies), 19 African countries (one study), Italy (one study), and Scotland (one historical cohort). There were no studies from

Fig. 1 Flow diagram for search results

Latin America, South Asia, or other regions experiencing extreme heat exposure. This narrow geographic scope, combined with differences in climatic conditions and socioeconomic contexts, limits generalizability of findings and highlights the lack of research on this topic.

Across studies, populations and developmental stages varied substantially, ranging from toddlers to adolescents and adults. Four studies examined prenatal heat exposure, while one focused on contemporaneous exposure in early childhood. Prenatal exposure was generally defined as deviations from local historical temperature norms, supporting within-location causal inference. Timing during gestation emerged as a potentially important dimension: the two China-based studies identified the first and second trimesters as periods of heightened sensitivity, whereas third-trimester exposure showed weaker or null associations. The single postnatal study (Italy) suggested that temperature and humidity may influence short-term emotional and

attentional states in toddlers, indicating that developmental timing may shape both mechanisms and outcomes.

Outcomes clustered into three domains: mental health, cognitive/educational outcomes, and behavioral/emotional functioning. Evidence for mental health outcomes was relatively consistent, with prenatal heat exposure associated with increased depressive or psychological distress symptoms in adolescence and adulthood. Cognitive and educational outcomes showed mixed findings: one large study in rural China documented adverse effects on schooling and literacy, while a study in mid-20th-century Scotland found no meaningful associations after accounting for seasonality and socioeconomic factors. Behavioral outcomes were assessed in only one small study, limiting inference regarding longer-term behavioral development.

Four studies explored effect modification, suggesting greater vulnerability among more recent birth cohorts, children in agriculturally dependent households, and those in

Table 1 Characteristics and findings of studies on prenatal or early childhood ambient heat exposure and developmental outcomes across the life course

Article	Sample Context	Method				Results		Limitations as reported by authors		
		Sample size	Age range (mean and SD, if specified)	Race, ethnicity, and/or urbanicity	Heat Exposure	Outcomes	Analytic approach		Data structure	
Adh-varyu et al. (2024)	Burkina Faso, Chad, Comoros, Republic of Congo, Côte d'Ivoire, Ethiopia, Ghana, Kenya, Malawi, Mali, Mauritania, Mauritius, Morocco, Namibia, Senegal, South Africa, Swaziland, Zambia, and Zimbabwe	61,885	18–65 years old	37% urban	Prenatal. Temperature in the calendar year before birth (resolution 0.5 degrees)	Mental health: Adult depression measured via Index for self-reported depression symptoms, diagnosis of depression in past 12 months, and medication for depression in past 2 weeks	Regression model with fixed effects for temperature point and year of birth. Additional covariates included rainfall the year before birth, gender and urbanicity	Cross sectional	A temperature increase of 1 °C in-utero increases self-reported symptoms of depression in adulthood by 0.03 SD. There is no consistent evidence for effects on diagnosis, treatment, or medication for depression	The effects are larger for the youngest (i.e., below median ages) respondents in the sample. It also assumes current residence reflects birth location, though robustness checks suggest this does not drive the results

Table 1 (continued)

Article	Sample Context	Method				Results		Limitations as reported by authors			
		Sample size	Age range (mean and SD, if specified)	Race, ethnicity, and/or urbanicity	Heat Exposure	Outcomes	Analytic approach		Data structure	Results	Moderation
Ai and Tan (2023)	China (126 cities in 25 provinces)	3,045	10–15 years old (M = 12.6; SD = 1.7)	59% Rural, 88% Han race	Prenatal. Multiple exposure variables, including average temperature during each pregnancy trimester and calculated deviations of number of days in each bin during pregnancy relative to historical average values	Mental health: Adolescent past month self-reported depression symptoms	Regression models with city of birth and year of birth fixed effects. Additional covariates included precipitation, relative humidity, wind speed, sunshine duration, air pressure, and demographic characteristics	Cross sectional	In utero exposure to one more day over 30 °C over typical temperature associated with increases in depression by 0.07 SD. Alternative models indicate that 1 °C warmer above average leads to increases by 0.16 SD. Associations between heat and decreased cognitive performance in test scores during adolescence	Effects are more pronounced in the first and second trimesters. Seasonal differences. Larger effects for adolescents born in the northern region and whose mothers worked in farming. Older adolescents also experienced stronger effects, No gender differences were found	The study is limited by the use of city-level averages and self-reported mental health measures, which may mask variation and introduce bias. It also assumes current city reflects birthplace, and estimated effects are small in magnitude
Ciucci et al. (2013)	Italy (Florence)	61	M=24.1 months (SD=3.6)	All children born in Italy. 8.5% of fathers and 18.6% of mothers came from Africa or South America	Early childhood. Air temperature was collected every 15 min from a close weather station in Center of Florence. Air temperature was in the classroom and in the garden was also collected using two weather sensors	Behavior and emotion: Daycare teachers' reports of children's behaviors and emotional states (Daily Behavioral and Emotional Questionnaire - DBEQ),	Multilevel regression, including gender, age, and time spent outdoors	Longitudinal	Temperature was not statistically significantly associated with attentional focusing, and was associated with aggression only in winter. No associations in the summer	Season moderated the weather-behavior association	The authors noted limitations, including the small, region-specific sample, the absence of data during autumn, reliance on teacher-rated measures, and the mild Mediterranean climate, which limits extrapolation to hotter environments

Table 1 (continued)

Article	Sample	Method			Outcomes	Analytic approach	Data structure	Results		Limitations as reported by authors
		Sample size	Age range (mean and SD, if specified)	Race, ethnicity, and/or urbanicity				Results	Moderation	
Hu and Li (2019)	China (25 provinces)	9,024	M = 36.96 years (SD = 10.41)	90% Han race; 100% rural-born	Language skills: Children's standardized word-test score on reading Chinese characters and literacy	Regression model including county-by-year, county-by-month, and year-by-month fixed effects. Additional covariates include other meteorological factors, gender, race, birth order, number of siblings, age at delivery, and parent education	Cross-sectional	Exposure to one additional high-temperature day during the in utero period was linked to a 0.18% higher risk of illiteracy and a 0.48% lower standardized word-test score	Larger associations for exposure in the second trimester of pregnancy. No differences according to quality of housing stock. Urban-individuals are less affected than rural	The study is limited by the lack of direct data on underlying mechanisms and by assuming county of birth reflects exposure despite possible migration. The authors also note that effects may be underestimated due to survival selection and limited adaptation
					Prenatal. Exposure to high temperatures during pregnancy. Number of days during gestational period with maximum temperatures above 85 °F	Regression models between temperature (linear) and outcomes including birth date (adjusting for seasonality)	Longitudinal	When controlling for the month of birth (i.e., seasonality), there are no associations between temperature and reading ability, arithmetic ability or teachers' assessment of behavioral problems	None	The study is limited by relying on temperature data from a single station, assuming uniform exposure, and by low climatic variation, which reduces power to detect small effects
Lawlor et al. (2006)	Scotland (Aberdeen)	12,150	Individuals born between 1950 and 1956. Cognitive assessments were conducted at ages 7, 9, and 11	Unclear	Cognition and behavior: Children's standardized test scores at ages 7, 9, and 11, including arithmetic and reading ability. Teachers also completed the Rutter scale of behavior problems	Prenatal. Mean outdoor temperature in the middle 10 days of each trimester of pregnancy, and mean temperature 10 days around conception				

All studies linked temperature via geographic location and date and therefore represent ecological or proxy rather than actual exposures

Table 2 Quality appraisal using the MMAT

	(1) Participants	(2) Measures	(3) Complete data	(4) Confounders	(5) Intervention
Adhvaryu et al. (2024)	Yes	Yes	Yes	Yes	Yes
Ai and Tan (2023)	Yes	Yes	No	Yes	Yes
Ciucci et al. (2013)	No	Yes	No	No	Yes
Hu and Li (2019)	Yes	Yes	No	Yes	Yes
Lawlor et al. (2006)	Yes	Yes	No	No	No

(1) Are the participants representative of the target population?; (2) Are measurements appropriate regarding both the outcome and intervention (or exposure)?; (3) Are there complete outcome data?; (4) Are the confounders accounted for in the design and analysis?; and (5) During the study period, is the intervention administered (or exposure occurred) as intended?

cooler regions experiencing atypical heat exposure. However, effect modifiers were examined inconsistently and within single studies, precluding cross-study conclusions. Overall, despite methodologically rigorous designs in at least three of the included large-scale studies, the small number of studies, limited geographic coverage, and heterogeneity in exposures and outcomes constrain the generalizability of findings. Taken together, the available evidence suggests that prenatal heat exposure may pose risks for mental health and, in some contexts, cognitive development, but substantial gaps remain regarding mechanisms, sensitive periods, and clinically meaningful impacts across diverse child and family contexts, and more research is urgently needed. Supplemental file S3 presents a detailed narrative summary of each study.

Quality Appraisal

Table 2 reports findings from the quality assessment. One study (Adhvaryu et al., 2024) met the five quality criteria, two studies using the same dataset (i.e., The China Family Panel Studies; Ai & Tan, 2023; Hu & Li, 2019), and the remaining two studies (Ciucci et al., 2013; Lawlor et al., 2006) did not meet three quality criteria, with small sample sizes and substantial confounding issues that limit the validity of their findings. The three studies that addressed selection bias employed fixed-effects models that accounted for baseline climatic conditions. All but one study used large sample sizes, some of which were nationally representative. All studies included measures with at least some evidence of reliability and validity. In most studies, however, outcome data were missing and not examined or addressed. These factors could have increased bias and should be considered when interpreting the observed associations.

Regarding certainty of the evidence, the use of MMAT as a guide to study quality indicated that the overall certainty of the evidence should be considered low. Only three studies employed quasi-experimental methods with adequate control of confounding, while two small observational studies had substantial risk of bias. The small number of studies, methodological heterogeneity, and limited direct replication

further reduce confidence in the estimated associations. Accordingly, the body of evidence provides suggestive but not conclusive support for an association between early-life heat exposure and later developmental outcomes.

Discussion

Our systematic review identified only five studies examining associations of prenatal and early childhood exposure to ambient heat with developmental outcomes across the life course. This small number contrasts with the growing body of conceptual and advocacy work (e.g., Early Childhood Scientific Council on Equity and the Environment, 2023; Rees, 2021) warning about the potential developmental risks of early exposure to global warming. While extensive research has documented links between ambient heat and children's physical health (Weeda et al., 2024), maternal, fetal, and newborn health (Chersich et al., 2020; Lakhoo et al., 2025), and adolescents' and adults' mental health and behavior (Ebi et al., 2021; Thompson et al., 2018), empirical evidence on developmental processes remains scarce. Understanding these effects empirically is essential for identifying which developmental outcomes are affected, through what mechanisms, for whom, and at what stages of development. Integrating developmental science with climate and environmental health research is crucial for moving beyond conceptualizing potential associations to understanding how early exposure to heat interacts with biological and psychosocial systems that shape developmental trajectories.

Of the five studies identified, four (three rated as high quality) reported associations between early-life exposure to high ambient temperatures and adverse cognitive, mental health, and behavioral outcomes, whereas one lower-quality study found null associations. Four studies also reported moderation effects by developmental stage, region, season, and urbanicity. These findings are broadly consistent with conceptual frameworks (Cuartas et al., 2024; Vergunst & Berry, 2022) proposing that excessive heat during sensitive developmental periods may directly and indirectly disrupt biological and psychosocial processes underlying

cognitive, socioemotional, and behavioral functioning, and that these effects will vary according to contextual vulnerabilities. However, the small number of studies, substantial methodological heterogeneity, and several study-level limitations preclude drawing firm conclusions about how prenatal and early childhood heat exposure influences lifelong development.

The reviewed studies differed markedly in their sampling strategies and in how they operationalized and quantified heat exposure, with implications for internal validity, interpretability, and generalizability. Three large-scale studies, one using data from 61,885 adult respondents in 19 African countries from the WHO World Health Surveys (Adhvaryu et al., 2024), and two based on the China Family Panel Studies (Ai & Tan, 2023; Hu & Li, 2019), linked birth dates and locations to high-resolution spatial temperature data. These studies focused on in utero exposure to above-average temperatures, leveraging geographic and temporal fixed effects to account for local climatic baselines and strengthen causal inference. In contrast, two small-sample studies from Italy (Ciucci et al., 2013) and Scotland (Lawlor et al., 2006) employed lower-resolution temperature data and weaker methodological designs, raising concerns about confounding and limiting causal interpretation. Consequently, most available evidence pertains to exposure to ambient heat in the prenatal period and to a few geographical regions (Africa, China, and Europe), leaving major gaps in Latin America, South and Southeast Asia, the Middle East, and other regions where young children face high exposure to heat-related hazards (Rees, 2021).

Developmental outcomes assessed across studies were also highly heterogeneous, limiting comparability and synthesis. Two studies examined adolescent and adult cognitive outcomes using standardized tests (Lawlor et al., 2006) or literacy and schooling measures (Hu & Li, 2019); others focused on teacher-reported behavioral ratings in childhood (Ciucci et al., 2013) and self-reported mental health in adulthood (Adhvaryu et al., 2024; Ai & Tan, 2023). This diversity in outcome domains, coupled with the absence of studies directly assessing early childhood development, hinders understanding of the specific developmental pathways through which early heat exposure exerts its effects. Notably, none of the reviewed studies examined socioemotional or executive function outcomes, skills that are foundational for later behavioral, cognitive, and mental health trajectories (Cunha & Heckman, 2007; Jones et al., 2016; Wolf & McCoy, 2019). Addressing these gaps is crucial for developing a comprehensive understanding of how early environmental stressors, such as heat, influence developmental processes across the life course.

Limitations of the Current Review

While this study makes an important contribution as the first systematic review to examine associations of prenatal and early childhood exposure to high ambient temperatures with developmental outcomes across the life course, it also has several limitations. First, we conducted a narrative rather than a meta-analytic synthesis, primarily due to the limited number and heterogeneity of available studies; therefore, we cannot draw conclusions about the average effect sizes reported in the literature. Second, although we did not exclude studies based on language, our search strategy relied on English-language keywords, and we may have inadvertently overlooked relevant studies published in other languages. Third, owing to data limitations, we were unable to thoroughly examine heterogeneity in the potential effects of early exposure to ambient heat on developmental outcomes, particularly differences between prenatal and postnatal exposure and across individual or contextual factors such as pre-existing vulnerabilities.

Future Directions for Research and Policy

The limitations acknowledged by the primary study authors, together with our review findings and quality appraisal, demonstrate the need for further interdisciplinary and globally representative research. Future work should clarify which developmental outcomes are affected by early-life exposure to high ambient temperatures, through what mechanisms, for whom, and at what developmental stages effects are most potent. Consistent with a geo-bioecological perspective (Cuartas & Morris-Perez, 2025), such research should consider mechanisms operating across multiple levels of the ecological model. A clearer distinction between direct and indirect exposures (e.g., direct heat stress vs. heat-related changes to caregiving and education) is also necessary and can help to more clearly organize the empirical literature. Although continued investigation of prenatal exposure and lifelong outcomes is critical, research should also focus on the early childhood period, a sensitive window for biological and psychosocial development (Berens & Nelson, 2019; Black et al., 2017; Draper et al., 2024).

Future studies should encompass a broader range of developmental domains, including cognitive, social, emotional, behavioral, and executive function skills, as well as mental health, to provide a comprehensive understanding of how excessive heat affects human development. To improve measurement precision and causal inference, research should integrate large-scale, representative developmental datasets with high-resolution temperature data and exploit quasi-experimental variation in temperature exposure. The use of longitudinal designs, alongside quasi-experimental

approaches applied to cross-sectional data, would further strengthen internal validity through the application of individual fixed-effects models. Additionally, greater attention is needed to identify mechanisms of transmission (e.g., pre-term birth, low birth weight) and moderating factors, which can inform the design of preventive and adaptive interventions. Importantly, such efforts should be conducted across diverse countries and contexts to distinguish universal from context-specific effects of early heat exposure.

Although the current evidence base is limited and somewhat inconclusive, emerging findings linking high ambient temperature with adverse developmental outcomes, together with broader evidence on the effects of excessive heat on the ecology of human development (Cuartas et al., 2024; Romanello et al., 2024; Vergunst & Berry, 2022), highlight the urgency of policy and intervention efforts to protect human development in a warming world. Ecological interventions that enhance preparedness, adaptation, and resilience amid rising temperatures are essential, including meteorological monitoring systems that inform early prevention measures. Targeted programs supporting pregnant women and young children, particularly those exposed to heatwaves or atypically high temperatures and those with pre-existing vulnerabilities such as lack of access to improved water and sanitation, healthcare, and education, among others, should be integrated into existing social protection, health, and education systems to safeguard children's health and developmental environments.

Conclusion

This systematic review reveals that empirical evidence on the developmental consequences of prenatal and early childhood exposure to high ambient temperatures remains limited, methodologically heterogeneous, and geographically narrow. Nonetheless, the available studies provide initial indications that early exposure to excessive heat may have lasting effects on cognitive, behavioral, and mental health outcomes. Expanding and strengthening this evidence base through interdisciplinary, longitudinal, and cross-contextual research is essential for advancing our understanding of how a warming climate shapes human development, and for informing interventions and policies that protect current and future generations from the developmental risks of a rapidly warming world.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10567-026-00569-3>.

Acknowledgements We thank Glenn Karlsen Bjerkenes, Head Librarian at the University of Oslo, for assisting us in developing the search strategy and conducting the searches.

Author Contributions Conceptualization: Cuartas, Narea, Ahun, Vergunst; Methodology: Cuartas, Narea, Ahun, Vergunst; Formal analysis and investigation: Cuartas, Narea, Vergunst; Writing - original draft preparation: Cuartas, Narea; Writing - review and editing: Cuartas, Narea, Ahun, Vergunst.

Funding None to declare.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Adhvaryu, A., Fenske, J., Kala, N., & Nyshadham, A. (2024). Fetal origins of mental health: Evidence from Africa. *Economic Development and Cultural Change*, 72(2), 493–515. <https://doi.org/10.1086/722536>
- Ai, H., & Tan, X. (2023). The effects of exposure to high temperatures during pregnancy on adolescent mental health: Evidence from China. *China Economic Review*, 80, 101991–101991. <https://doi.org/10.1016/j.chieco.2023.101991>
- Berens, A. E., & Nelson, C. A. (2019). Neurobiology of fetal and infant development: Implications for infant mental health. In C. H. Zeanah (Ed.), *Handbook of Infant Mental Health*. Guilford Press.
- Black, M., Walker, S. P., Fernald, L. C. H., Andersen, C. T., DiGirolamo, A. M., Lu, C., McCoy, D. C., Fink, G., Shawar, Y. R., Shiffman, J., Deverece, A. E., Wodon, Q. T., Vargas-Barón, E., & Grantham-McGregor, S. (2017). Early childhood development coming of age: Science through the life course. *Lancet*, 389(10064), 77–90. [https://doi.org/10.1016/S0140-6736\(16\)31389-7](https://doi.org/10.1016/S0140-6736(16)31389-7)
- Bronfenbrenner, U. (2005). *Making human beings human: Bioecological perspectives on human development*. Sage Publications Ltd.
- Bronfenbrenner, U., & Morris, P. (2006). The bioecological model of human development. In W. Damon, R. M. Lerner, & R. M. Lerner (Eds.), *Handbook of Child Psychology* (6 ed., pp. 793–828). Wiley. <https://doi.org/10.1002/9780470147658.chpsy0114>
- Burke, M., Hsiang, S. M., & Miguel, E. (2015). Climate and conflict. *Annual Review of Economics*, 7(1), 577–617. <https://doi.org/10.1146/annurev-economics-080614-115430>
- Chersich, M. F., Pham, M. D., Areal, A., Haghighi, M. M., Manyuchi, A., Swift, C. P., Wernecke, B., Robinson, M., Hetem, R., Boeckmann, M., & Hajat, S. (2020). Associations between high temperatures in pregnancy and risk of preterm birth, low birth weight, and stillbirths: Systematic review and meta-analysis. *BMJ*, 371, Article m3811. <https://doi.org/10.1136/bmj.m3811>
- Choi, H. M., Heo, S., Foo, D., Song, Y., Stewart, R., Son, J., & Bell, M. L. (2024). Temperature, crime, and violence: A systematic review and meta-analysis. *Environmental Health Perspectives*, 132(10), Article 106001. <https://doi.org/10.1289/EHP14300>
- Ciucci, E., Calussi, P., Menesini, E., Mattei, A., Petralli, M., & Orlandini, S. (2013). Seasonal variation, weather and behavior in day-care children: A multilevel approach. *International Journal of Biometeorology*, 57(6), 845–856. <https://doi.org/10.1007/s00484-012-0612-0>
- Cuartas, J., & Camacho, A. (2025). High temperatures and violent child punishment at home: Evidence from six countries. *Psychology of Violence*. <https://doi.org/10.1037/vio0000651>

- Cuartas, J., McCoy, D. C., Torres, I., Burghardt, L., Shonkoff, J. P., & Yoshikawa, H. (2024). The developmental consequences of early exposure to climate change-related risks. *Child Development Perspectives*, 18(3), 145–154. <https://doi.org/10.1111/cdep.12503>
- Cuartas, J., & Morris-Perez, P. (2025). Bringing natural ecosystems into the bioecological model of human development. *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-025-02334-y>
- Cuartas, J., & Vergunst, F. (2025). Nurturing care as a critical buffer against climate change impacts on child development. *PLOS Global Public Health*, 5(4), Article e0004441. <https://doi.org/10.1371/journal.pgph.0004441>
- Cunha, F., & Heckman, J. (2007). The technology of skill formation. *American Economic Review*, 97(2), 31–47. <https://doi.org/10.1257/aer.97.2.31>
- Dekkers, O. M., Vandenbroucke, J. P., Cevallos, M., Renehan, A. G., Altman, D. G., & Egger, M. (2019). COSMOS-E: Guidance on conducting systematic reviews and meta-analyses of observational studies of etiology. *PLoS Medicine*, 16(2), Article e1002742. <https://doi.org/10.1371/journal.pmed.1002742>
- Draper, C. E., Yousafzai, A. K., McCoy, D. C., Cuartas, J., Obradović, J., Bhopal, S., Fisher, J., Jeong, J., Klingberg, S., Milner, K., Pisani, L., Roy, A., Seiden, J., Sudfeld, C. R., Wrottesley, S. V., Fink, G., Nores, M., Tremblay, M. S., & Okely, A. D. (2024). The next 1000 days: Building on early investments for the health and development of young children. *Lancet*. [https://doi.org/10.1016/S0140-6736\(24\)01389-8](https://doi.org/10.1016/S0140-6736(24)01389-8)
- Early Childhood Scientific Council on Equity and the Environment. (2023). *Extreme Heat Affects Early Childhood Development and Health: Working Paper No. 1*. www.developingchild.harvard.edu
- Ebi, K. L., Capon, A., Berry, P., Broderick, C., de Dear, R., Havenith, G., Honda, Y., Kovats, R. S., Ma, W., Malik, A., Morris, N. B., Nybo, L., Seneviratne, S. I., Vanos, J., & Jay, O. (2021). Hot weather and heat extremes: Health risks. *Lancet*, 398(10301), 698–708. [https://doi.org/10.1016/S0140-6736\(21\)01208-3](https://doi.org/10.1016/S0140-6736(21)01208-3)
- Goto, R., Frodl, T., & Skokauskas, N. (2021). Armed conflict and early childhood development in 12 low- and middle-income countries. *Pediatrics*. <https://doi.org/10.1542/peds.2021-050332>
- Haddaway, N. R., Grainger, M. J., & Gray, C. T. (2021). *citation-chaser: An R package for forward and backward citations chasing in academic searching*. In (Version 0.0.3) <https://github.com/nealhaddaway/citationchaser>
- Hong, Q. N., Pluye, P., Fàbregues, S., Bartlett, G., Boardman, F., Cargo, M., Dagenais, P., Gagnon, M.-P., Griffiths, F., & Nicolau, B. (2018). Mixed methods appraisal tool (MMAT), version 2018. *Registration of copyright*, 1148552(10), 1–7.
- Hsiang, S. M., Burke, M., & Miguel, E. (2013). Quantifying the influence of climate on human conflict. *Science*, 341(6151), Article 1235367. <https://doi.org/10.1126/science.1235367>
- Hu, Z., & Li, T. (2019). Too hot to handle: The effects of high temperatures during pregnancy on adult welfare outcomes. *Journal of Environmental Economics and Management*, 94, 236–253. <https://doi.org/10.1016/j.jeem.2019.01.006>
- IPCC. (2022). *Climate change 2022: Impacts, adaptation and vulnerability*. Cambridge University Press. <https://doi.org/10.1017/9781009325844>
- IPCC. (2023). *Climate Change 2023: Synthesis Report*. IPCC, Geneva, Switzerland: Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)].
- Jones, S. M., Zaslow, M., Darling-Churchill, K. E., & Halle, T. G. (2016). Assessing early childhood social and emotional development: Key conceptual and measurement issues. *Journal of Applied Developmental Psychology*, 45, 42–48. <https://doi.org/10.1016/j.appdev.2016.02.008>
- Kadir, A., Shenoda, S., & Goldhagen, J. (2019). Effects of armed conflict on child health and development: A systematic review. *PLoS One*, 14(1), Article e0210071. <https://doi.org/10.1371/journal.pone.0210071>
- Kiarsi, M., Amiresmaili, M., Mahmoodi, M. R., Farahmandnia, H., Nakhiae, N., Zareiyan, A., & Aghababaeian, H. (2023). Heat waves and adaptation: A global systematic review. *Journal of Thermal Biology*, 116, Article 103588. <https://doi.org/10.1016/j.jtherbio.2023.103588>
- Koepp, A. E., Lanza, K., Byrd-Williams, C., Bryan, A. E., & Gershoff, E. T. (2023). Ambient temperature increases and preschoolers' outdoor physical activity. *JAMA Pediatrics*, 177(5), 539–540. <https://doi.org/10.1001/jamapediatrics.2023.0067>
- Kohler, T. A., Smith, M. E., Bogaard, A., Feinman, G. M., Peterson, C. E., Betzenhauser, A., Pailles, M., Stone, E. C., Marie Prentiss, A., Dennehy, T. J., Ellyson, L. J., Nicholas, L. M., Fauseit, R. K., Styring, A., Whitlam, J., Fochesato, M., Foor, T. A., & Bowles, S. (2017). Greater post-Neolithic wealth disparities in Eurasia than in North America and Mesoamerica. *Nature*, 551(7682), 619–622. <https://doi.org/10.1038/nature24646>
- Lakhoo, D. P., Brink, N., Radebe, L., Craig, M. H., Pham, M. D., Haghighi, M. M., Wise, A., Solarin, I., Luchters, S., Maimela, G., Chersich, M. F., Chikandiwa, A., Nakstad, B., Wright, C. Y., Brimicombe, C., Technau, K.-G., Harden, L., Boeckmann, M., Strehlau, R., Group, H. H. S. (2025). A systematic review and meta-analysis of heat exposure impacts on maternal, fetal and neonatal health. *Nature Medicine*, 31(2), 684–694. <https://doi.org/10.1038/s41591-024-03395-8>
- Lawlor, D. A., Clark, H., Ronalds, G., & Leon, D. A. (2006). Season of birth and childhood intelligence: Findings from the Aberdeen Children of the 1950s cohort study. *British Journal of Educational Psychology*, 76(3), 481–499. <https://doi.org/10.1348/000709905X49700>
- Levy, B. S., Sidel, V. W., & Patz, J. A. (2017). Climate change and collective violence. *Annual Review of Public Health*, 38(1), 241–257. <https://doi.org/10.1146/annurev-publhealth-031816-044232>
- Magnan, A. K., Pörtner, H. O., Duvat, V. K. E., Garschagen, M., Guinder, V. A., Zommers, Z., Hoegh-Guldberg, O., & Gattuso, J. P. (2021). Estimating the global risk of anthropogenic climate change. *Nature Climate Change*, 11(10), 879–885. <https://doi.org/10.1038/s41558-021-01156-w>
- Miles-Novelo, A., & Anderson, C. A. (2022). *Climate Change and Human Behavior: Impacts of a Rapidly Changing Climate on Human Aggression and Violence*. Cambridge University Press. <https://doi.org/10.1017/9781108953078>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ (Clinical research ed.)*, 372, Article n71. <https://doi.org/10.1136/bmj.n71>
- Prentice, C. M., Meneveau, M., Wu, Y., Girard, L. C., Zheng, Y., & Vergunst, F. (2025). Heat effects on children's education outcomes: A systematic review and meta-analysis. *PsyArXiv*. <https://doi.org/10.31234/osf.io/6jrx3>
- Rees, N. (2021). The Climate Crisis Is a Child Rights Crisis: Introducing the Children's Climate Risk Index. *UNICEF*.
- Romanello, M., Walawender, M., Hsu, S. C., Moskeland, A., Palmeiro-Silva, Y., Scamman, D., Ali, Z., Ameli, N., Angelova, D., Ayeb-Karlsson, S., Basart, S., Beagley, J., Beggs, P. J., Blanco-Villafuerte, L., Cai, W., Callaghan, M., Campbell-Lendrum, D., Chambers, J. D., Chicmana-Zapata, V., & Costello, A. (2024). The 2024 report of the lancet countdown on health and climate change: Facing record-breaking threats from delayed action.

- Lancet*, 404(10465), 1847–1896. [https://doi.org/10.1016/S0140-6736\(24\)01822-1](https://doi.org/10.1016/S0140-6736(24)01822-1)
- Sanson, A. V., Malca, K. V. P., & Van Hoorn, J. (2022). *Impact of the Climate Crisis on Children's Social Development* (pp. 206–223). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781119679028.ch11>
- Sanson, A. V., Van Hoorn, J., & Burke, S. E. L. (2019). Responding to the impacts of the climate crisis on children and youth. *Child Development Perspectives*, 13(4), 201–207. <https://doi.org/10.1111/cdep.12342>
- Sapolsky, R. M. (2017). *Behave: The biology of humans at our best and worst*. Penguin.
- Thompson, R., Hornigold, R., Page, L., & Waite, T. (2018). Associations between high ambient temperatures and heat waves with mental health outcomes: A systematic review. *Public Health*, 161, 171–191. <https://doi.org/10.1016/j.puhe.2018.06.008>
- Thompson, R., Lawrance, E. L., Roberts, L. F., Grailey, K., Ashrafian, H., Maheswaran, H., Toledano, M. B., & Darzi, A. (2023). Ambient temperature and mental health: A systematic review and meta-analysis. *Lancet Planetary Health*, 7(7), e580–e589. [https://doi.org/10.1016/S2542-5196\(23\)00104-3](https://doi.org/10.1016/S2542-5196(23)00104-3)
- Vergunst, F., & Berry, H. L. (2022). Climate change and children's mental health: A developmental perspective. *Clinical Psychological Science*, 10(4), 767–785. <https://doi.org/10.1177/21677026211040787>
- Walker, S., Wachs, T. D., Grantham-McGregor, S., Black, M. M., Nelson, C. A., Huffman, S. L., Baker-Henningham, H., Chang, S. M., Hamadani, J. D., Lozoff, B., Gardner, J. M. M., Powell, C. A., Rahman, A., & Richter, L. (2011). Inequality in early childhood: Risk and protective factors for early child development. *Lancet*, 378(9799), 1325–1338. [https://doi.org/10.1016/S0140-6736\(11\)60555-2](https://doi.org/10.1016/S0140-6736(11)60555-2)
- Weeda, L. J. Z., Bradshaw, C. J. A., Judge, M. A., Saraswati, C. M., & Le Souëf, P. N. (2024). How climate change degrades child health: A systematic review and meta-analysis. *Science of the Total Environment*, 920, Article 170944. <https://doi.org/10.1016/j.scitotenv.2024.170944>
- Wolf, S., & McCoy, D. C. (2019). The role of executive function and social-emotional skills in the development of literacy and numeracy during preschool: A cross-lagged longitudinal study. *Developmental Science*, 22(4), Article e12800. <https://doi.org/10.1111/desc.12800>

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